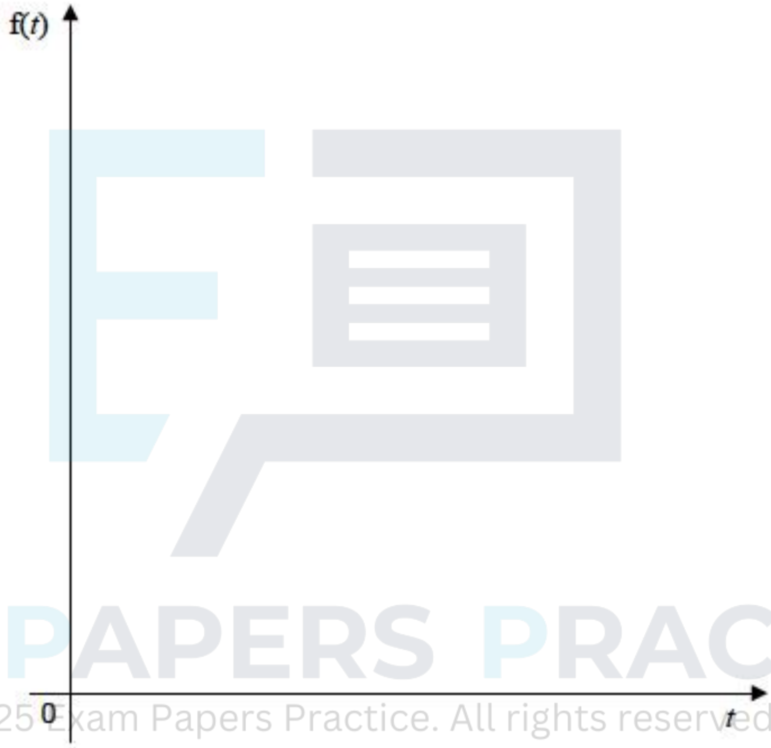
Q1.

The continuous random variable T has probability density function defined by

$$f(t) = \begin{cases} \dfrac{1}{3} & 0 \le t \le \dfrac{3}{2} \\ \dfrac{9-2t}{18} & \dfrac{3}{2} \le t \le \dfrac{9}{2} \\ 0 & \text{otherwise} \end{cases}$$

(a) (i) Sketch this probability density function below.



(2)

(ii) State the median of T.

(1)

(b) (i) Find E(T)

(2)

(ii) Given that $E(T^2) = \dfrac{15}{4}$, find Var $(4T - 5)$

(3)
(Total 8 marks)

Q2.

A continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} kx^2 & 0 \leqslant x \leqslant 3 \\ 9k & 3 \leqslant x \leqslant 4 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f.

(3)

(b) Show that the value of k is $\frac{1}{18}$.

(4)

(c) (i) Write down the median value of X.

(ii) Calculate the value of the lower quartile of X.

(4)

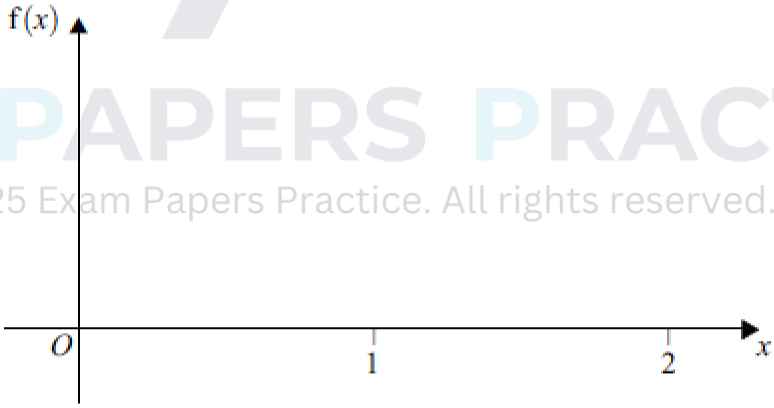
(Total 11 marks)

Q3.

A continuous random variable X has the probability density function defined by

$$f(x) = \begin{cases} x^2 & 0 \leqslant x \leqslant 1 \\ \frac{1}{3}(5 - 2x) & 1 \leqslant x \leqslant 2 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f on the axes below.



(3)

(b) (i) Find the cumulative distribution function, F, for $0 \leq x \leq 1$.

(2)

(ii) Hence, or otherwise, find the value of the lower quartile of X.

(2)

(c) (i) Show that the cumulative distribution function for $1 \leq x \leq 2$ is defined by

$$F(x) = \frac{1}{3}(5x - x^2 - 3)$$

(4)

(ii) Hence, or otherwise, find the value of the upper quartile of X.

(4)
(Total 15 marks)

Q4.

The random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{40}(x+7) & 1 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f.

(2)

(b) Find the **exact** value of $E(X)$.

(3)

(c) Prove that the distribution function F, for $1 \leq x \leq 5$, is defined by

$$F(x) = \frac{1}{80}(x + 15)(x - 1)$$

(4)

(d) Hence, or otherwise:

(i) find $P(2.5 \leq X \leq 4.5)$;

(2)

(ii) show that the median, m, of X satisfies the equation $m^2 + 14m - 55 = 0$.

(3)

(e) Calculate the value of the median of X, giving your answer to three decimal places.

(2)
(Total 16 marks)

Q5.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{3}{8}x^2 & 0 \leq x \leq \frac{1}{2} \\ \frac{3}{32} & \frac{1}{2} \leq x \leq 11 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f.

(3)

(b) Show that:

(i) $P(X \geq 8\frac{1}{3}) = \frac{1}{4}$

(ii) $P(X \geq 3) = \frac{3}{4}$.

(3)

(c) Hence write down the **exact** value of:

(i) the interquartile range of X;

(ii) the median, m, of X.

(3)

(d) Find the **exact** value of $P(X < m \mid X \geq 3)$.

(3)

(Total 12 marks)

Q6.

The continuous random variable X has the probability density function defined by

$$F(x) = \begin{cases} \frac{3}{8}(x^2+1) & 0 \leq x \leq 1 \\ \frac{1}{4}(5-2x) & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

(a) The cumulative distribution function of X is denoted by $F(x)$.

Show that, for $0 \leq x \leq 1$,

$$F(x) = \frac{1}{8}x(x^2+3)$$

(3)

(b) Hence, or otherwise, verify that the median value of X is 1.

(2)

(c) Show that the upper quartile, q, satisfies the equation $q^2 - 5q + 5 = 0$ and

hence that $q = \frac{1}{2}(5-\sqrt{5})$.

(5)

(d) Calculate the **exact** value of $P(q < X < 1.5)$.

(4)

(Total 14 marks)

Q7.

The random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{2} & 0 \le x \le 1 \\ \frac{1}{8}(x-4)^2 & 1 \le x \le 4 \\ 0 & \text{otherwise} \end{cases}$$

(a) State values for the median and the lower quartile of X.

(2)

(b) Show that, for $1 \le x \le 4$, the cumulative distribution function, $F(x)$, of X is given by

$$F(x) = 1 + \frac{1}{54}(x-4)^3$$

(You may assume that $\int (x-4)^2 \, dx = \frac{1}{3}(x-4)^3 + c$.)

(4)

(c) Determine $P(2 \le X \le 3)$.

(2)

(d) (i) Show that q, the upper quartile of X, satisfies the equation $(q - 4)^3 = -13.5$.

(3)

(ii) Hence evaluate q to three decimal places.

(1)

(Total 12 marks)

Q8.

The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{1}{2} & 0 \le x \le 1 \\ \frac{3-x}{4} & 1 \le x \le 3 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f.

(3)

(b) Explain why the value of η, the median of X, is 1.

(2)

(c) Show that the value of μ, the mean of X, is $\frac{13}{12}$.

(4)

(d) Find P($X < 3\mu - \eta$).

(3)
(Total 12 marks)

Q9.

The continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & x < -1 \\ \dfrac{x+1}{k+1} & -1 \leq x \leq k \\ 1 & x > k \end{cases}$$

where k is a positive constant.

(a) Find, in terms of k, an expression for P($X < 0$).

(2)

(b) Determine an expression, in terms of k, for the lower quartile, q_1.

(3)

(c) Show that the probability density function of X is defined by

$$f(x) = \begin{cases} \dfrac{1}{k+1} & -1 \leq x \leq k \\ 0 & \text{otherwise} \end{cases}$$

(2)

(d) Given that $k = 11$:

(i) sketch the graph of f;

(2)

(ii) determine E(X) and Var(X);

(2)

(iii) show that P($q_1 < X <$ E(X)) = 0.25.

(2)
(Total 13 marks)

Q10.

Engineering work on the railway network causes an increase in the journey time of commuters travelling into work each morning.

The increase in journey time, T hours, is modelled by a continuous random variable with probability density function

$$f(t) = \begin{cases} 4t(1-t^2) & 0 \leqslant t \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Show that $E(T) = \frac{8}{15}$.

(3)

(b) (i) Find the cumulative distribution function, $F(t)$, for $0 \leq t \leq 1$.

(2)

(ii) Hence, or otherwise, for a commuter selected at random, find

P(mean < T < median)

(5)

(Total 10 marks)

Q11.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{5}(2x+1) & 0 \leqslant x \leqslant 1 \\ \frac{1}{15}(4-x)^2 & 1 < x \leqslant 4 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f.

(2)

(b) (i) Show that the cumulative distribution function, $F(x)$, for $0 \leq x \leq 1$ is

$F(x) = \frac{1}{5}x(x+1)$

(3)

(ii) Hence find $P(X \geq 0.5)$.

(2)

(iii) Verify that the lower quartile, q_1, satisfies

$0.5 < q_1 < 0.75$

(3)

(Total 10 marks)

Q12.

The continuous random variable X has probability density function defined by

$$f(x) = \begin{cases} \frac{1}{5}(2x+1) & 0 \le x \le 1 \\ \frac{1}{15}(4-x)^2 & 1 < x \le 4 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f.

(2)

(b) (i) Show that the cumulative distribution function, F(x), for $0 \le x \le 1$ is

$$F(x) = \frac{1}{5}x(x+1)$$

(3)

(ii) Hence write down the value of P($X \le 1$).

(1)

(iii) Find the value of x for which P($X \ge x$) = $\frac{17}{20}$.

(5)

(iv) Find the lower quartile of the distribution.

(4)

(Total 15 marks)

Q13.

The time, T hours, that the supporters of Bracken Football Club have to queue in order to obtain their Cup Final tickets has the following probability density function.

$$f(t) = \begin{cases} \frac{1}{5} & 0 \le t < 3 \\ \frac{1}{45}t(6-t) & 3 \le t \le 6 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f.

(3)

(b) Write down the value of P(T = 3).

(1)

(c) Find the probability that a randomly selected supporter has to queue for at least 3 hours in order to obtain tickets.

(2)

(d) Show that the median queuing time is 2.5 hours.

(2)

(e) Calculate P(median < T < mean).

(6)

(Total 14 marks)

Q14.

The error, X millimetres, made when the heights of prospective members of a new gym club are measured can be modelled by a rectangular distribution with the following probability density function.

$$f(x) = \begin{cases} k & -4 \le x \le 6 \\ 0 & \text{otherwise} \end{cases}$$

(a) State the value of k.

(1)

(b) Write down the value of E(X).

(1)

(c) Calculate P(X > 0).

(2)

(d) The height of a randomly selected prospective member is measured. Find the probability that the **magnitude** of the error made exceeds 3.5 millimetres.

(3)
(Total 7 marks)

Q15.

The discrete random variable X has

E(X) = 4 and E(X^2) = 17.2.

(a) Find the value of Var (X).

(2)

(b) A circle has radius (X + 8).

Find, in terms of π, values for the mean and variance of the **circumference**, C, of the circle.

(4)

(c) The **area**, S, of the circle, with radius (X + 8), is to be expressed in the form

$$\pi(X^2 + aX + b),$$

where a and b are positive constants.

(i) Find values for a and b.

(2)

(ii) Hence find, in terms of π, the value of E(S).

(2)
(Total 10 marks)

Q16.

At a Post Office, the time, T minutes, that customers have to wait in order to be served has the following probability density function.

$$f(t) = \begin{cases} \dfrac{t^2}{18} & 0 \le t < 3 \\ \dfrac{1}{4}(5 - t) & 3 \le t \le 5 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Write down the value of P(T = 4).

(1)

(b) Show that the median waiting time is 3 minutes.

(2)

(c) Find the probability that customers have to wait for less than 3.5 minutes to be served.

(5)

(d) Calculate the mean time that customers have to wait in order to be served.

(5)

(Total 13 marks)

Q17.

All consultations with an optician are by appointment. The time, T minutes by which an appointment is delayed has the following probability density function.

$$f(t) = \begin{cases} \frac{1}{54}t^2 & 0 \le t < 3 \\ \frac{1}{6} & 3 \le t < 6 \\ \frac{1}{24}(10 - t) & 6 \le t < 10 \\ 0 & \text{otherwise} \end{cases}$$

(a) Sketch the graph of f.

(4)

(b) Use your graph to find:

(i) P(3 ≤ T < 6);

(2)

(ii) P(T > 6);

(2)

(iii) the lower quartile

(4)

(c) What proportion of appointments is delayed by less than 2 minutes?

(4)

(Total 16 marks)

Q18.

The delay, T minutes, in connecting electronically to a ticket booking service can be modelled by the distribution function:

$$P(T \le t) = 1 - e^{-2t}, \qquad t \ge 0.$$

(a) (i) Write down the probability density function of T.

(1)

(ii) State values for the mean and standard deviation of T.

(2)

(b) Calculate the probability that the delay exceeds 3 minutes.

(2)

(c) Find the median delay, giving your answer in seconds, to the nearest second.

(3)

(Total 8 marks)

Q19.

The total amount of added time, S minutes, played per match by a team in a local football league is distributed with probability density function given by:

$$f(s) = \begin{cases} ks^2 & 1 \le s \le 5 \\ 0 & \text{otherwise} \end{cases}$$

(a) Show that the exact value of the constant k is $\frac{3}{124}$.

(4)

(b) Determine $P(S < 3)$.

(3)

(c) Calculate the median amount of added time played per match, giving your answer to the nearest minute.

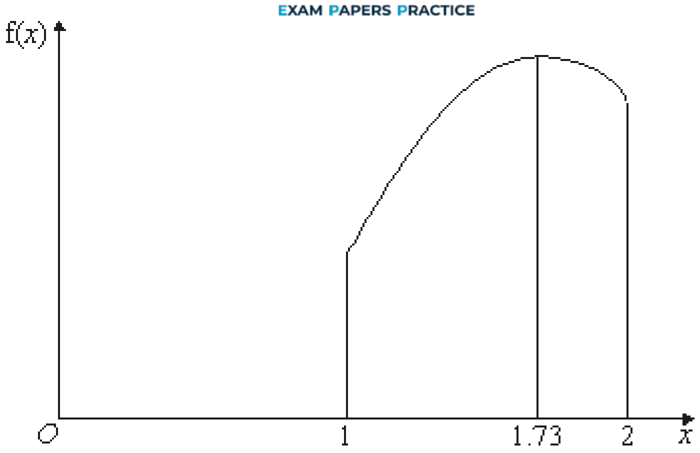
(4)

(Total 11 marks)

Q20.

The sketch below shows the probability density function $f(x)$ of the continuous random variable X.

($f(x) = 0$ outside the interval $1 \le x \le 2$.)



You are given that $\int_1^{1.56} f(x)\,dx = 0.5$ $\quad$ $\int_{1.73}^{2} f(x)\,dx = 0.291$

$\int_1^2 x f(x)\,dx = 1.54$ $\quad$ $\int_1^2 (x - 1.54)^2 f(x)\,dx = 0.0751$

(a) Write down the value of:

(i) $\int_1^2 f(x)\,dx$;

(ii) the mean of X;

(iii) the median of X;

(iv) the mode of X;

(v) the range of X.

(5)

(b) Find:

(i) the probability that X lies between the median and the mode;

(2)

(ii) the standard deviation of X;

(2)

(iii) E(X^2).

(2)

(Total 11 marks)

Q21.

The random variable, X, has probability density function

$$f(x) = \begin{cases} cx + d & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases} \qquad \text{where } c \text{ and } d \text{ are constants}$$

(a) Show that $2c + 2d = 1$.

(3)

(b) Find the median of X:

(i) when $c = 0$ and $d = 0.5$;

(2)

(ii) when $c = 0.5$ and $d = 0$.

(3)

(c) The following pairs of values are suggested for c and d.

Suggestion A: $c = 0.5$ and $d = 0.5$
Suggestion B: $c = -0.5$ and $d = 1$
Suggestion C: $c = 1.5$ and $d = -1$

Only one of the suggestions provides a valid probability density function.

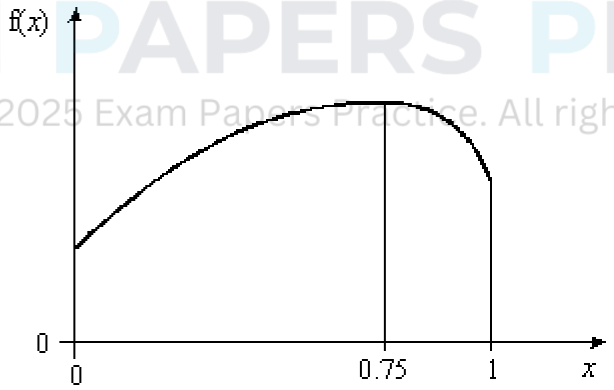
For **each** suggestion state whether or not it is valid. If you state it is not valid, give a reason.

(5)
(Total 13 marks)

Q22.

The sketch below shows the probability density function f(x) of the continuous random variable X.

[f(x) = 0 outside the interval $0 \le x \le 1$]



(a) Explain why the value of the mode of X is 0.75.

(1)

(b) You are given that:

$E(X) = 0.542$ $E(X^2) = 0.369$

$\int_0^{0.557} f(x)dx = 0.500$ $\int_0^{0.542} f(x)dx = 0.483$ $\int_0^{0.75} f(x)dx = 0.719$

(i) Find the standard deviation of X.

(3)

(ii) Find the probability that X exceeds the mode.

(2)

(iii) Write down the value of the median of X.

(1)

(iv) Evaluate the probability that X lies between the median and the mean.

(3)

(Total 10 marks)